SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA1604	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL2, BL3,BTL4)	**Assessment:**	Assessment Tool 1 – Short Answer Test
Weightage:		**Total Marks:**	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	VINO ALDRIN E	**Reg. No.:**	192511481
Section / Batch:	Slot - D / IInd year	**Date:**	03/10/26

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
	GRAND TOTAL:		**100 Marks**

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.
(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.
(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct (192511481)

I VINO ALDRIN E *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: E Vinoaldrin

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
					Overall Total (100)		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

03/10/26

Assessment Tool - 1

Short Answer Test

CO 1

Name :- VINO ALDRIN E
Roll No. :- 192511481
Course Code :- CSA1604

Q.1. **City Hospital Data Warehouse :-**

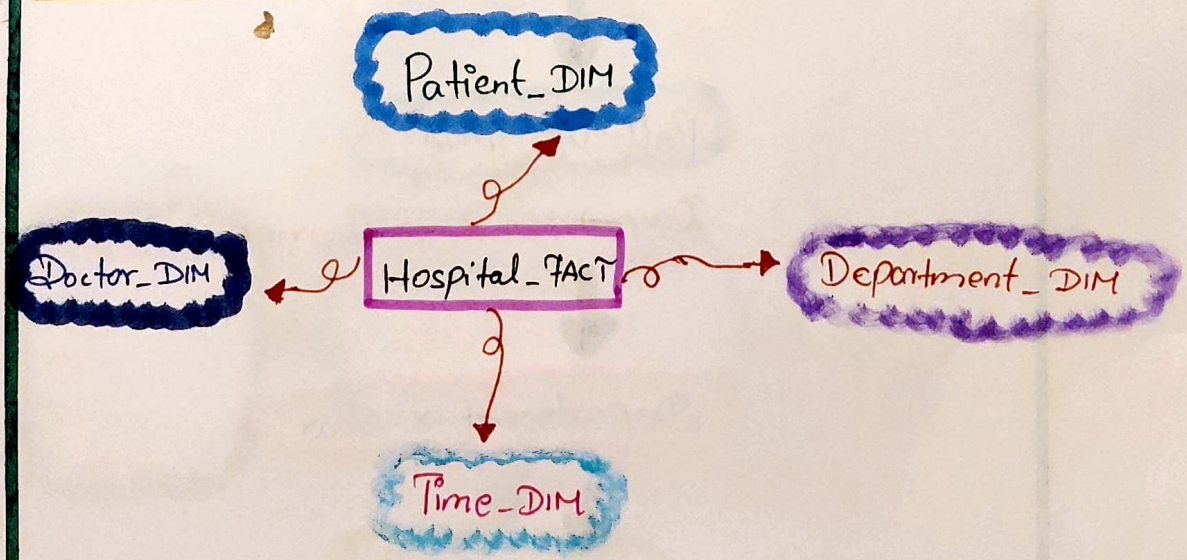
Given :-

Dimensions : Patient, Doctor, Department, Time.

Measures : Visit_Count, avg_charge.

- At lowest level → actual Consultation fee.
- At higher levels → average Consultation fee.

(a) **Snowflake Schema :-**

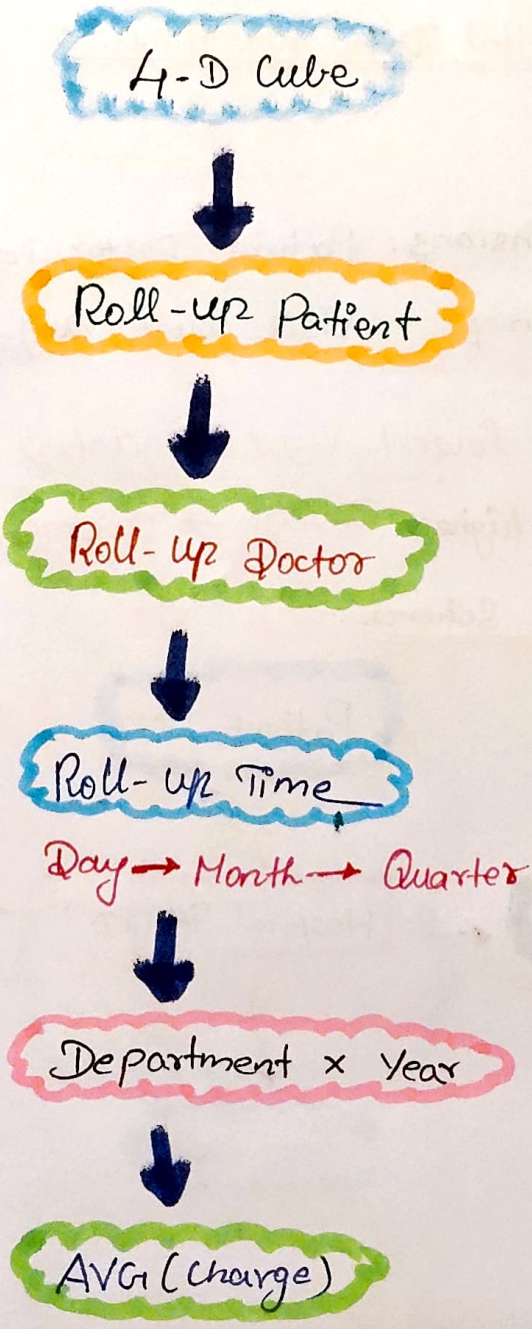


(b) OLAP Operations :-

Starting Cube:

Patient x Doctor x Department x Time

To obtain average consultation fee department-wise for each year :-



(c) Number of Cuboids :-

Each dimension has 4-levels.

Formula :-

$$\text{Number of Cuboids} = (L_1+1)(L_2+1)(L_3+1)(L_4+1)$$

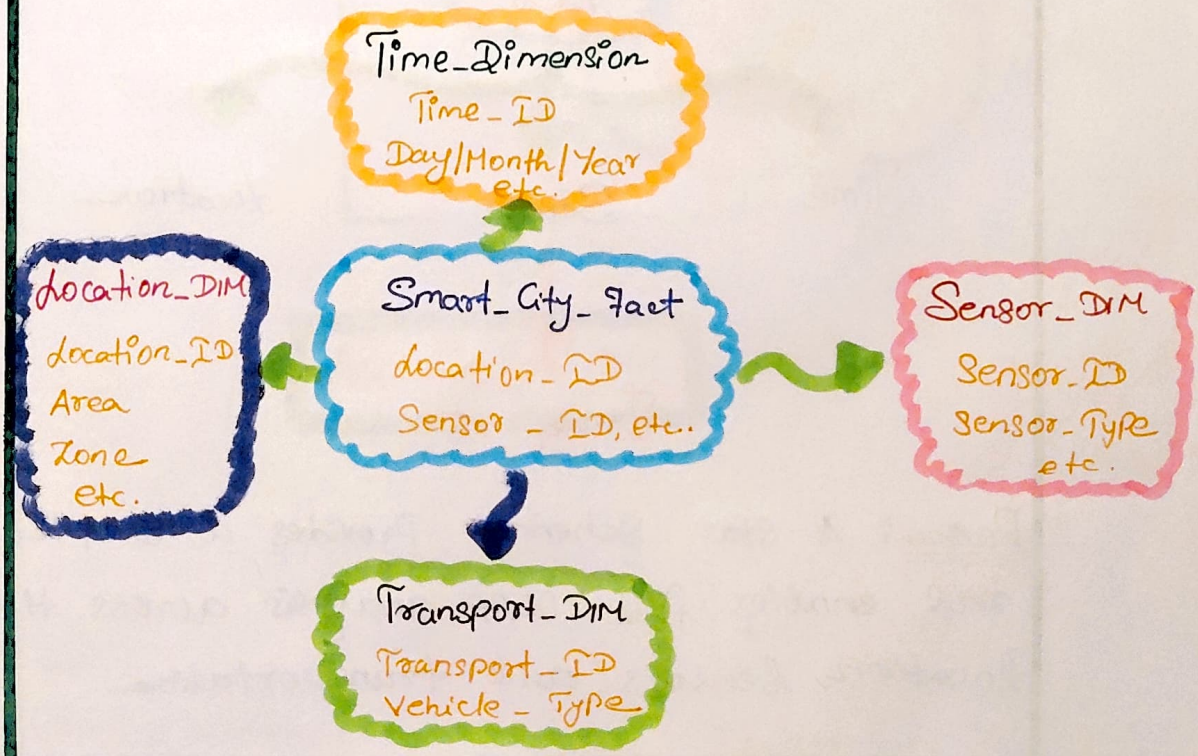
$$= (4+1)^4$$

$$= \boxed{625}$$

∴ Answer = 625 Cuboids.

Q.2. Smart City Data-Warehouse :-

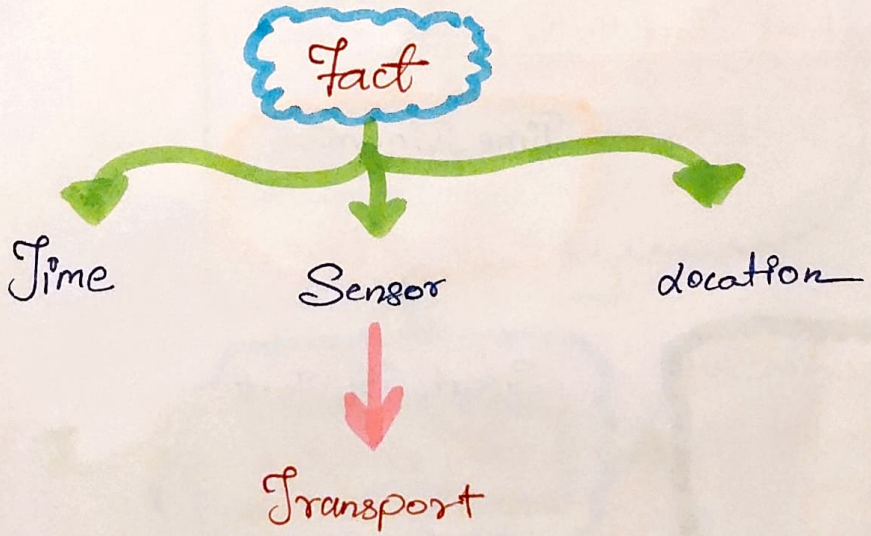
Central Fact Table :-



Dimensions :-

Dimension	Important Attributes
Time	Day, month, quarter, year
Location	Area, Zone, City
Sensor	Type, Source
Transport	Vehicle type, route.
Citizen	Citizen ID, Feedback

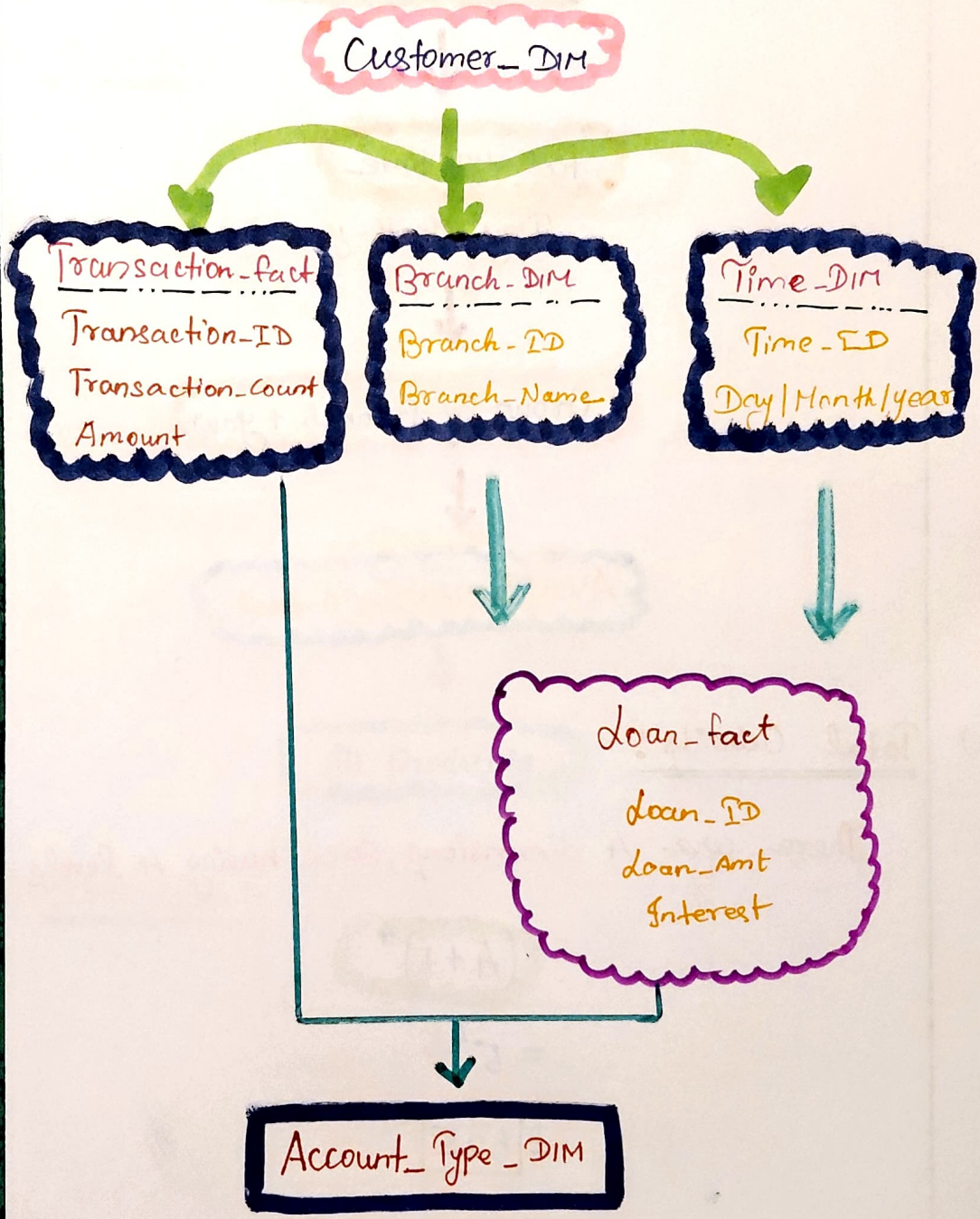
Why Star Schema?



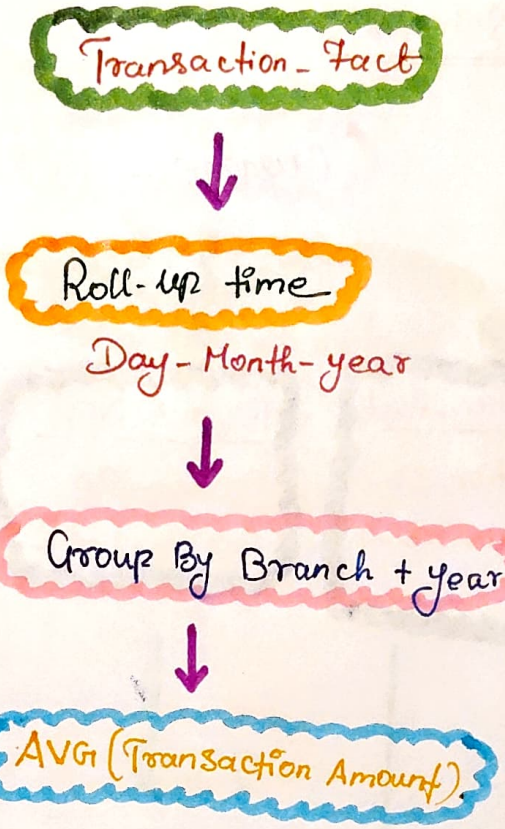
Reason: A star schema provides a simple structure and enables fast OLAP analysis across time, location, sensors and transportation.

Q.3. Banking Data Warehouse – Galaxy Schema.

(a) Galaxy Schema:-



(b) OLAP Operation :-



(c) Total Cuboids :-

There are 4 dimensions, each having 4 levels :-

$$(4+1)^4$$

$$= 5^4$$

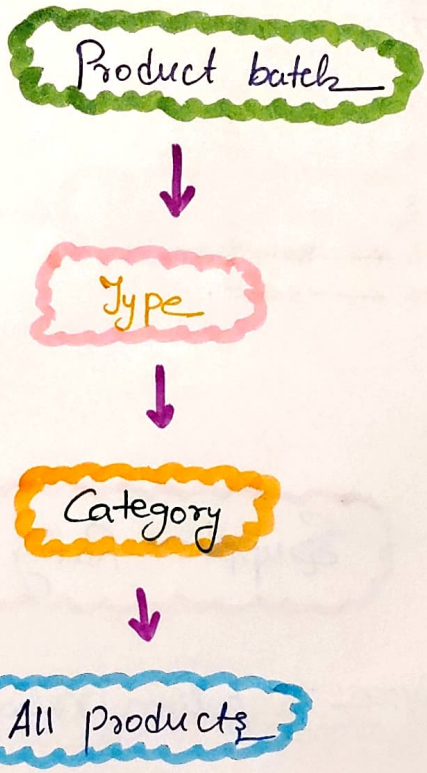
$$= \boxed{625}$$

Since this is a galaxy schema with two fact tables, the same shared dimensional cuboid space is applicable to analyse each fact table.

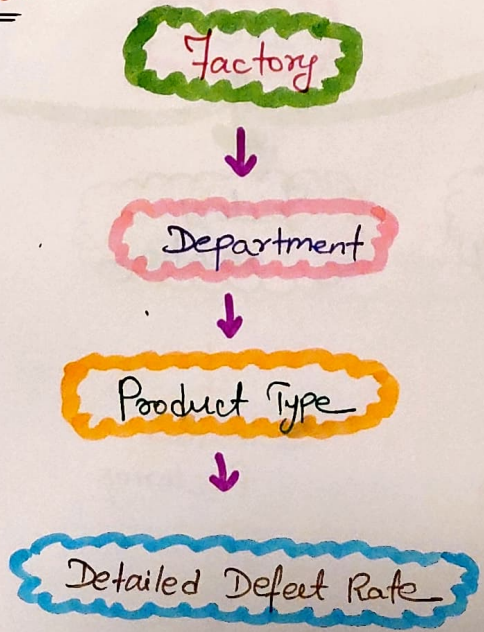
Q.4. Manufacturing Data Warehouse :-

OLAP Operations :-

1. Roll-up :-

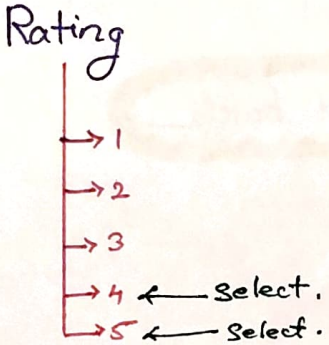


2. Drill down :-



3. Slice :-

Supplier Dimension -

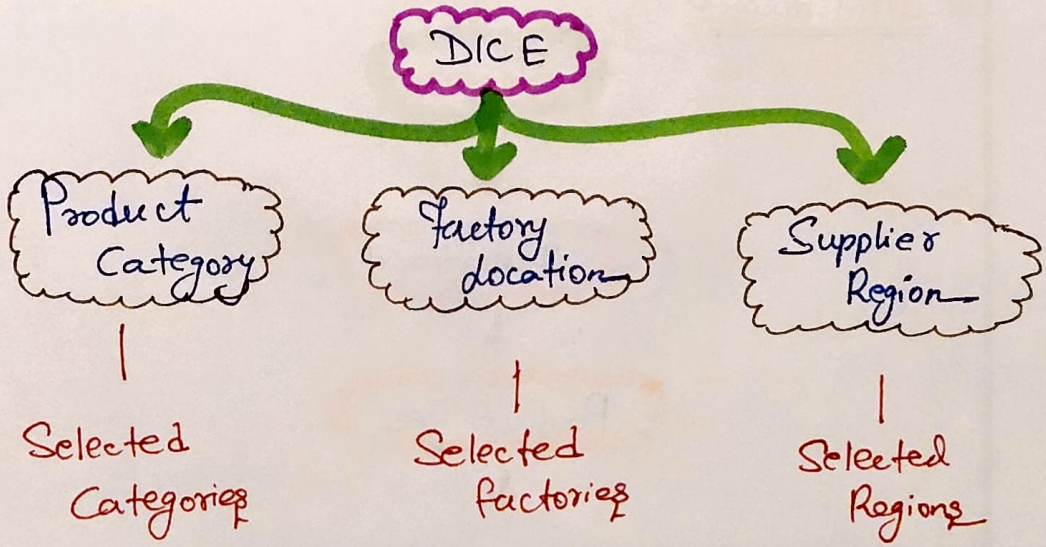


Condition :-

Supplier Rating ≥ 4

* Then analyze Supplier reliability.

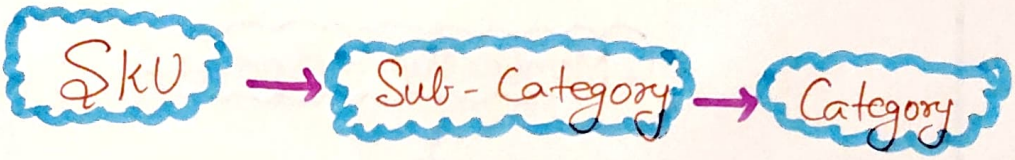
4. Dice :-



Q.5. Retail Chain :-

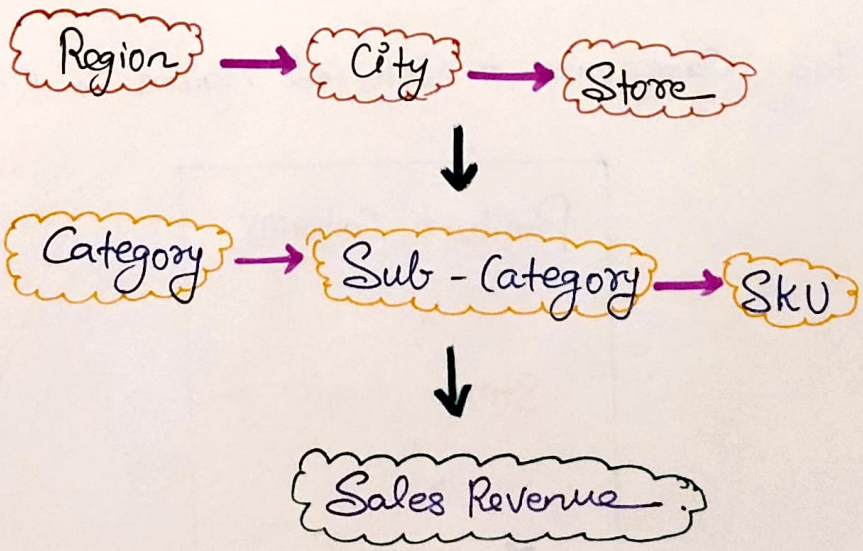
1. Roll-up :

Inventory turnover by product category, last quarter, all stores.



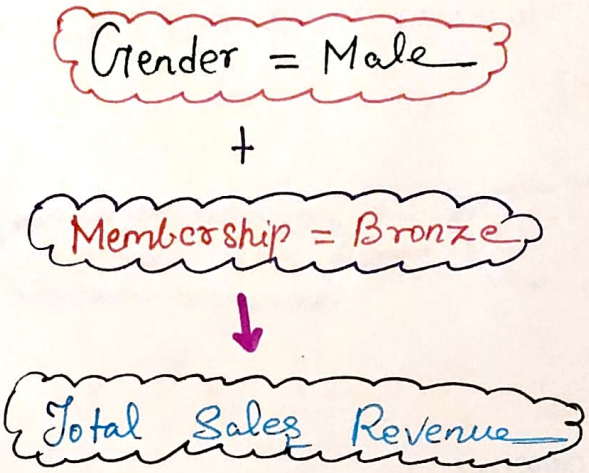
2. Drill-down :-

Sales by SKU in a specific region.



3. Slice :-

Male + Bronze Customers across all Stores.



4. Dice :-

Selected Categories + Selected regions + Selected Periods.

